DiaTwin: A Circadian-Aware, Clinician-in-the-Loop Platform for Personalised Insulin Dosing in Type 1 Diabetes

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Introduction

Over 8 million people with Type 1 Diabetes (T1D) worldwide rely on multiple daily injections (MDI) with insulin pens. Despite increasing CGM adoption, no affordable decision-support tool simultaneously integrates insulin pharmacodynamics, macronutrient-specific meal absorption, physical activity effects, and circadian variation in insulin sensitivity. Fixed insulin-to-carbohydrate ratios (ICR) and insulin sensitivity factors (ISF) prescribed by clinicians cannot adapt to these dynamics, producing chronic dosing errors, hypoglycaemia risk, and significant cognitive burden. This work presents DiaTwin, a modular digital health platform designed to address these gaps within a clinician-supervised framework.

Methods

DiaTwin integrates four physiological layers into a single on-device dose recommendation: (1) camera-based meal recognition with gamma-distribution postprandial absorption curves for full macronutrient loads (CHO, protein, fat), referenced to the UVA/Padova GI subsystem; (2) insulin-on-board (IOB) and meal-on-board (MOB) tracking via analogue-specific PK/PD parameters per the AWMF S3 Guideline (2024); (3) a three-state ODE activity model producing real-time Activity-on-Board (AOB) and an 8-hour post-exercise ISF/ICR sensitivity correction; and (4) a circadian baseline anchor using per-2-hour ISF/ICR blocks derived from CGM calibration or finger-prick data. All recommendations are explainable and clinician-reviewable under a clinician-in-the-loop paradigm, with all meal-insulin-activity-outcome tuples logged for future reinforcement learning (RL) parameter refinement. The platform engine was pre-trained on and validated against the AZT1D dataset (25 T1D patients with CGM, bolus, and carbohydrate records, spanning both MDI and automated insulin delivery periods).

Results

The platform demonstrated strong technical performance, ingesting 50,000 database entries in 0.88 seconds and sustaining stable operation across 4,738 real-data processing cycles. The modular architecture successfully integrated all four physiological subsystems into a single unified recommendation pipeline, with dose suggestions rounded to 0.1 U precision and a 0.5 U safety floor. Clinical expert review by a board-certified diabetologist confirmed the platform's physiological plausibility and highlighted the significant unmet clinical need for integrating blood glucose, insulin dosage, physical activity, and nutritional data into a single decision-support application. The clinician further recognised the system's learning architecture as valuable for both patient education and physician-informed decision making.

Conclusion

DiaTwin demonstrates that circadian-aware, multi-context insulin dosing support is technically feasible for MDI users without requiring automated insulin delivery hardware. The platform provides a cost-accessible, equity-oriented, and ML-ready foundation for advancing personalised T1D pharmacotherapy. Future work will focus on a prospective pilot RCT (N=30, 6 months) validating circadian RL-driven adaptation against standard-of-care MDI therapy, with particular attention to time-in-range improvement and hypoglycaemia reduction in both high-resource and low-resource clinical settings.